

THAPAR INSTITUTE OF ENGINEERING AND TECHNOLOGY

Patiala, Punjab — India

VISITOR MANAGEMENT SYSTEM

Multi-Gate Digital Entry & Exit Platform

Complete Product Requirements & System Design Document

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TABLE OF CONTENTS

This document contains the following sections:

1. Executive Overview
2. Problem Statement
3. Objectives
4. Scope
5. Stakeholders
6. User Roles & Permissions Matrix
7. Assumptions
8. Constraints
9. Risks & Recommended Corrections
10. Functional Requirements
11. Non-Functional Requirements
12. Visitor Category Definitions & Field Tables
13. Detailed End-to-End User Flows
14. Gate-wise Queue Workflow
15. Entry Workflow
16. Exit Workflow
17. Exception & Edge Case Handling
18. Recommended Approval Rules
19. Search & Retrieval Rules
20. Guard Shift / Gate Assignment Logic
21. Fast Approval / Fast Exit Features
22. Dashboard Requirements
23. Reports & CSV Export Requirements
24. Alerting & Notification Requirements
25. Authentication & Authorization Design
26. Audit Trail Design
27. Data Model / Database Schema
28. Suggested API Endpoints
29. Screen-by-Screen UI Module Breakdown
30. Admin Control Features
31. Supervisor Features
32. Guard Features
33. Operational Policies
34. Privacy, Security & Data Retention Analysis
35. Abuse / Fraud / Misuse Prevention
36. Scalability Recommendations
37. Future Enhancements & AI Features
38. Recommended Tech Stack
39. MVP vs Phase-2 Feature Split
40. Testing Strategy
41. Deployment Considerations
42. Final Product Recommendations
43. Deliverables Summary (A–J)
44. Product Summary & Build Order
45. Best Prompt for Claude Code Generation
46. Unresolved Questions

SECTION 1 — EXECUTIVE OVERVIEW

Thapar Institute of Engineering and Technology (TIET) operates a sprawling residential campus in Patiala, Punjab, served by four entry/exit gates. The current visitor management process is entirely manual: paper registers at gates, no cross-gate visibility, no searchable records, and no systematic overstay tracking. Security personnel have no digital tooling to verify identities, correlate entries with exits, or escalate concerns.

This document defines the complete design blueprint for a Visitor Management System (VMS) — a multi-platform, role-aware digital platform that transforms gate operations from paper-based guesswork to real-time, accountable, searchable, and auditable digital workflows.

Platform Summary

Dimension	Specification
Platform	Web App (primary) + Mobile-responsive interface
Gates Covered	Front Gate, Western Gate, Polytechnical Gate, TSLAS Gate
User Roles	Gate Guard · Supervisor · Head Admin
Visitor Categories	Parent · Delivery · Taxi · Vendor · Contractor · Event Guest · Maintenance · Emergency · Interview Candidate · Courier · Alumni · Other Services
Key Capabilities	QR-triggered intake · Selfie capture · Guard queue · Approve/Reject/Escalate · Entry/Exit logging · Overstay alerts · Full audit trail · CSV export
Phase 1 MVP	Core entry/exit flow, guard panel, basic admin dashboard, QR intake
Phase 2	AI face match, analytics engine, watchlist, repeat visitor history, emergency mode

SECTION 2 — PROBLEM STATEMENT

Current State

Thapar University's gate operations suffer from the following documented pain points:

- Paper registers are inconsistently maintained and not searchable.
- There is no cross-gate coordination — a visitor entering at Front Gate cannot be located at other gates.
- Guard accountability is absent — no record of which guard approved or rejected a visitor.
- Overstay situations are unknown unless physically observed.
- No photo verification — any name can be written on paper.
- Exit reconciliation is manual, slow, and error-prone.
- No central visibility for supervisors or administration.
- Delivery and taxi flows are undifferentiated from visiting flows.
- Emergency situations have no fast-track or escalation mechanism.

Desired Future State

A digital, gate-specific, role-aware platform where:

1. Every visitor self-registers via QR scan at the gate kiosk.
2. A guard reviews and approves or rejects the entry in real time.
3. Entry and exit are logged with guard, gate, and timestamp.
4. Any guard at any gate can search and mark exit for a visitor.
5. Supervisors and head admins have cross-gate visibility and control.
6. An overstay alert fires when a visitor exceeds expected or maximum duration.
7. Head admin has full, tamper-proof audit capability.

SECTION 3 — OBJECTIVES

Primary Objectives

8. Replace paper-based gate registers with a fully digital, searchable system.
9. Enable real-time visitor queue management per gate.
10. Enforce selfie-based photo identity verification for all non-standard visitor types.
11. Provide guards with fast, mobile-friendly approve/reject/escalate tools.
12. Enable cross-gate exit — visitor enters Gate A, exits Gate B.
13. Log every action with guard identity, gate, role, and timestamp.
14. Alert guards and supervisors when visitors overstay.
15. Provide supervisors and admin with live dashboards and exportable reports.
16. Maintain a full, immutable audit trail accessible only to Head Admin.
17. Design for real-world gate conditions: intermittent mobile connectivity, rushed guards, and high-volume peak hours.

Secondary Objectives

- Build a blacklist/watchlist system for repeat offenders.
- Provide repeat visitor history for faster future check-ins.
- Enable gate-load analytics for shift planning.
- Design for future AI-powered face matching (Phase 2).

SECTION 4 — SCOPE

In Scope

- Visitor self-registration via QR code triggered web form.
- Guard panel for queue management, verification, approve/reject/escalate.
- Entry and exit logging across all 4 gates.
- Cross-gate exit search and marking.
- Supervisor multi-gate view and intervention.
- Head Admin full-system dashboard, audit trail, user management.
- Overstay alerting and notification to guards.
- Reports, CSV export, gate-wise and category-wise analytics.
- Authentication, RBAC, session management, audit trail.
- Blacklist/watchlist management (MVP-light, full in Phase 2).

Out of Scope

- Student or hostel warden portal — not in scope.
- Automated student database lookup — not available; manual entry only.
- Parent-side pre-approval chain — not required.
- Integration with university ERP or HRMS — out of scope for Phase 1.
- Physical hardware (barrier gates, RFID, biometric) — hardware not in scope.
- Payment or fine collection.
- Vehicle access control beyond logging.

SECTION 5 — STAKEHOLDERS

Stakeholder	Role in System	Key Needs
Head Admin	System owner, full control	Full audit, user management, all-gate visibility, security override
Supervisor	Multi-gate oversight	Cross-gate queue view, escalation handling, report generation
Gate Guard	Front-line operator	Fast queue processing, approve/reject/escalate, exit search
Visitor (indirect)	Data submitter only	Simple QR-based form, clear status, fast processing
University Security Dept	Policy owner	Compliance, data security, shift accountability
IT / Dev Team	Builders & maintainers	Clear API design, clean schema, deployable stack

SECTION 6 — USER ROLES & PERMISSIONS MATRIX

Legend:  = Full Access  = Limited/Own Gate Only  = No Access

Action / Feature	Guard 	Supervisor 	Head Admin 	Notes
Submit visitor form (visitor side)	N/A	N/A	N/A	Public-facing, no auth needed
View own gate queue				Guard sees own gate only
View all gate queues				Cross-gate supervisory view
Approve visitor entry				Guard approves own gate; Supervisor can override
Reject visitor entry				Reason mandatory
Escalate to Supervisor				Guard-only escalation trigger
Handle escalated cases				Supervisor resolves escalation
Mark visitor exit				Any gate guard can mark exit
Search active visitors				All roles search across gates
View visitor selfie				Identity verification
Add to blacklist				Supervisor can flag; Admin confirms
Remove from blacklist				Admin only
Override approved entry				With mandatory reason + audit
Cancel active visit				Supervisor+ only
Manage guard accounts				Admin only
Manage supervisor accounts				Admin only
Reset guard password				Supervisor can reset own team
View full audit trail				Head Admin only
Export CSV reports				Supervisor exports own gate; Admin all
View all-gate live dashboard				Supervisors see all gates
Configure visitor categories				Admin only
Configure overstay alert rules				Admin only
Emergency mode activation				Supervisor+ can trigger
Gate assignment (self)				Guard sets own gate on login

Shift reports				Guard views own; Supervisor all
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SECTION 7 — ASSUMPTIONS

- Guards have basic smartphone or desktop access at their gate posts.
- Each gate has a stable enough data connection to run the web application. Resilience logic is still designed in, but hard offline mode is not a Phase 1 requirement.
- No student database is available or accessible. All student-related fields (name, hostel, year) are manually entered by visitors and manually verified by guards. Validation is social/visual, not automated.
- QR codes are physically printed and posted at each gate. Each gate has its own unique QR code embedding the gate ID.
- The primary submission surface is a mobile browser on the visitor's phone. No native app download is required from the visitor's side.
- Guards log in with credentials pre-provisioned by Admin. Self-registration of guards is not permitted.
- One guard is on duty per gate at any time, though multiple guards can be logged into the same gate concurrently during handovers.
- Head Admin is a single senior security official or a small trusted team.
- All visitor submissions are in English. Multi-language support is not in Phase 1 scope.
- Photo storage infrastructure (cloud or on-premise) is to be provisioned by the university IT team.

SECTION 8 — CONSTRAINTS

- No integration with any external student, faculty, or visitor database available.
- Guards must be able to process a visitor in under 90 seconds to avoid gate queues.
- The visitor form must be completable on a phone without requiring account creation or app download.
- Selfie capture must work on standard mobile browsers without requiring special plugins.
- The system must not store raw biometric data; photos are identification aids, not biometric identifiers for legal purposes.
- All role escalation stops at Head Admin — no external routing.
- The system is internal to the university network or deployed on a secured domain. Public exposure of admin panels is not permitted.

SECTION 9 — RISKS & RECOMMENDED CORRECTIONS

R1 — Selfie Storage Forever

 **RISK/FLAG:** Storing selfies indefinitely creates GDPR/PDPB (India's Personal Data Protection Bill) liability, massive storage growth, and a high-value target for data breaches.

RECOMMENDED SAFER POLICY: Retain selfie images for 12 months from capture date. After 12 months, replace the image with a cryptographic hash of its content (SHA-256) to prove the image existed without storing identifiable data. Implement a scheduled job (nightly) to enforce this. Provide Head Admin with a manual retention extension option for flagged cases (e.g., ongoing investigation).

 **RECOMMENDED DEFAULT:** 12-month active retention → hash-only after expiry. Configurable by Admin.

R2 — No Student Database Validation

 **RISK/FLAG:** Visitors can type any student name, hostel, and year. A malicious visitor could impersonate being associated with a real student.

MITIGATION: (a) Guards are trained to call the student if visiting hours or circumstances seem suspicious. (b) A 'Call Student to Verify' flag is available as a guard option. (c) Phase 2 introduces optional SMS verification to student's registered number. (d) Guards should cross-check hostel name against known hostel list (admin-configurable).

R3 — Guard Queue Tampering

 **RISK/FLAG:** If guards can edit or delete queue items, bad actors could remove visit records.

MITIGATION: Guards cannot delete records. They can only change status (approve/reject/escalate). All status changes are logged with guard ID, timestamp, and IP. Only Head Admin can void a visit, and voiding creates an audit entry, not a deletion.

R4 — Overstay Alert Logic (Original: 12h + 12h Warning)

 **RISK/FLAG:** The original requirement states 'if inside 12h, alert that person needs to exit in next 12h.' This means a person can stay 24h before anything happens — operationally odd for most visitor types.

RECOMMENDED OVERSTAY MODEL: Each visitor category has a configurable expected stay duration. During form fill, visitors provide an expected exit time. Alert tiers: (1) 30 minutes before expected exit → notify guard. (2) At expected exit time if not yet exited → escalate alert to supervisor. (3) 4 hours past expected exit → critical alert to supervisor + head admin. (4) Hard overstay cap: if no expected time given, use category default (e.g., Delivery = 2h, Parent = 4h, Taxi = 1h). Alerts stop once the visitor exits.

 **RECOMMENDED DEFAULT:** Category-based expected duration defaults with three alert tiers: reminder → escalation → critical.

R5 — Cross-Gate Exit Without Entry Record

 **RISK/FLAG:** What if exit is marked but no matching entry record exists? Could indicate system error, manual override, or gate bypass.

MITIGATION: Exit marking requires selecting an active inside record. If no matching record is found, guard can create an 'unknown exit' log with required explanation. This is logged as anomaly in audit trail.

R6 — Duplicate Active Sessions

 **RISK/FLAG:** A visitor might submit the form multiple times (e.g., slow form, refreshed page). Multiple pending requests for same person could confuse guards.

MITIGATION: Deduplicate on phone number + gate + within 10-minute window. If a submission is made within 10 minutes of an existing pending/approved record at the same gate, show 'You have already submitted — status: X' rather than creating duplicate.

R7 — Guard Account Compromise

 **RISK/FLAG:** If a guard account is shared or credentials leaked, false approvals could be made.

MITIGATION: Sessions are short-lived (4h). Login requires gate selection at each session. Concurrent logins from different IPs trigger supervisor alert. Failed login lockout after 5 attempts.

SECTION 10 — FUNCTIONAL REQUIREMENTS

FR-01: Visitor Self-Registration

- Visitor scans gate-specific QR code which opens a mobile-optimised web form.
- Form pre-populates the gate name based on QR code parameter.
- Visitor selects a category from a defined list.
- Category-specific fields are shown dynamically.
- Selfie capture is mandatory for: Parent, Other Services, Vendor, Contractor, Event Guest, Interview Candidate, Alumni.
- Selfie is optional but encouraged for Delivery and Taxi.
- Phone number is mandatory for all categories.
- On submission, a Submission Reference ID is shown to the visitor.

FR-02: Guard Queue Management

- Guard sees a real-time queue of pending visitors for their assigned gate.
- Each queue item shows: visitor name, category, submission time, selfie thumbnail.
- Guard can open a full detail view per visitor.
- Guard actions: Approve, Reject (with reason), Escalate (with reason).
- Guard can use Quick Approve button for returning visitors or short checks.
- Queue auto-refreshes every 10 seconds. Manual refresh option available.
- New item notification (sound/vibration on mobile, badge count on desktop).

FR-03: Entry Logging

- On approval, an Entry record is created with: visitor_id, guard_id, gate_id, entry_timestamp, guard name.
- Visitor status changes to 'entered'.
- Entry record is visible in the active-inside tracker.

FR-04: Exit Workflow

- Any guard at any gate can initiate an exit search.
- Search supported by: name, phone, vehicle number, category, company name, submission reference ID.
- Search returns active (entered, not exited) visitor records.
- Guard selects the correct visitor and clicks Mark Exit.
- Exit record stores: guard_id, exit_gate_id, exit_timestamp.
- Visit status changes to 'exited'.
- If exit gate differs from entry gate, this is flagged in report but allowed.

FR-05: Escalation Flow

- Guard can escalate a pending or approved visit to Supervisor.
- Escalation requires a reason (free text, mandatory).
- Supervisor sees escalated items highlighted in their multi-gate queue view.
- Supervisor can: Approve, Reject, Hold, or Return to Guard.
- All escalation actions are logged with actor and timestamp.

FR-06: Overstay Alerting

- System checks active visitors every 5 minutes against expected exit time.
- Alert tiers: Reminder (–30 min) → Escalation (at exit time) → Critical (+4h over).
- Alerts displayed in guard panel, supervisor dashboard.
- Push notification (or in-app alert) sent to on-duty guard for that gate.

FR-07: Blacklist / Watchlist

- Supervisor can flag a visitor for watchlist review.
- Head Admin can promote a flagged visitor to blacklist.
- On future form submission, system checks phone number and ID against blacklist.
- If blacklisted, guard is alerted with a red warning on the queue item.
- Blacklisted entries are not auto-rejected; guard must make a decision.

FR-08: Admin & Supervisor Controls

- Head Admin manages guard and supervisor accounts.
- Head Admin configures category-specific overstay thresholds.
- Head Admin views full audit trail with filters.
- Head Admin can export any report as CSV.
- Supervisor manages their gate assignment for subordinate guards.
- Supervisor can override guard decisions with mandatory reason.

SECTION 11 — NON-FUNCTIONAL REQUIREMENTS

Category	Requirement
Performance	Form submission → Guard queue appearance ≤ 2 seconds. Guard approve action → status update ≤ 1.5 seconds.
Availability	99.5% uptime target. Scheduled maintenance only during 2:00–4:00 AM IST.
Scalability	Handle 500 concurrent visitors across 4 gates. Support burst of 100 submissions in 10 minutes during events.
Mobile UX	Visitor form fully functional on mobile browsers (Chrome Android, Safari iOS). No app install required.
Image Handling	Selfies compressed to max 800KB before storage. Max upload size 5MB with client-side resizing.
Security	HTTPS everywhere. JWT-based auth. XSS/CSRF/SQL injection protections. Rate limiting on form submission.
Audit Integrity	Audit logs are append-only. No delete capability. Stored separately from operational DB.
Accessibility	WCAG 2.1 AA for all guard and admin panels. High-contrast mode for outdoor use.
Browser Support	Latest 2 versions of Chrome, Firefox, Safari, Edge. IE not supported.
Resilience	Queue state maintained in server session. Guard panel shows last-known queue on connectivity loss with stale indicator.
Localisation	English only in Phase 1. Architecture must support i18n for Phase 2 (Punjabi/Hindi).
Data Backup	Daily automated DB backups. 30-day backup retention. Point-in-time recovery for last 7 days.

SECTION 12 — VISITOR CATEGORY DEFINITIONS & FIELD TABLES

CAT-01: Parent / Family Member

Purpose: Family visiting a hostel student. Highest volume category on weekends.

Selfie: Mandatory

Field Name	Type	Required	Notes
Visitor Full Name	Text	Yes	Primary identification
Visitor Phone Number	Phone	Yes	For deduplication and future contact
Government ID Type	Dropdown	Yes	Aadhaar / Passport / DL / Voter ID
Government ID Number	Text	Yes	Last 4 digits only stored; full number displayed to guard only
Student Name (to visit)	Text	Yes	Manual entry; no DB validation
Student Hostel Name	Dropdown	Yes	Admin-configurable hostel list
Student Current Year	Dropdown	Yes	1st / 2nd / 3rd / 4th / 5th Year
Student Contact Number	Phone	No	Optional; for guard to call and verify
Vehicle Number	Text	No	If arriving by own vehicle
Number of Accompanying Persons	Number	No	For group visits
Purpose of Visit	Text	No	Brief free-text reason
Selfie	Image	Yes	Live capture preferred; upload fallback
Expected Exit Time	Time	Yes	Used for overstay alert

 **NOTE:** Guard should visually compare selfie to visitor's face. No automated matching in Phase 1.

CAT-02: Delivery Person

Purpose: Courier, food delivery, package delivery for students or departments.

Selfie: Recommended (not mandatory)

Field Name	Type	Required	Notes
Delivery Company / App Name	Text	Yes	E.g., Amazon, Zomato, Swiggy, DTDC
Delivery Person Name	Text	Yes	
Delivery Person Phone	Phone	Yes	
Recipient Name	Text	Yes	Student or department receiving the delivery
Recipient Hostel / Building	Text	Yes	Where to deliver
Vehicle Number	Text	No	Required if arriving by vehicle
Expected Exit Time	Time	Yes	Delivery visits should be short; default max 45 min

Package Description	Text	No	Brief description of package
Tracking Number / Order ID	Text	No	Optional reference for verification

 **NOTE:** Recommended Default: If expected exit time not provided, system sets default overstay cap to 45 minutes for delivery category.

CAT-03: Taxi / Cab Driver

Purpose: Pickup or drop service for students, staff, or guests. Key distinction from delivery: taxi waits or drops, delivery delivers goods.

Selfie: Optional

Field Name	Type	Required	Notes
Driver Name	Text	Yes	
Driver Phone	Phone	Yes	
Cab Company / App	Text	No	E.g., Ola, Uber, local cab; solo driver if private
Vehicle Number	Text	Yes	Mandatory for taxi — critical identification
Vehicle Type	Dropdown	Yes	Auto / Sedan / SUV / Mini-bus
Visit Type	Dropdown	Yes	Drop-off / Pickup / Both
Passenger Name (if known)	Text	No	Person being dropped or picked up
Expected Exit Time	Time	Yes	Required; Recommended Default: 30 min
Selfie	Image	No	Optional for taxi

 **NOTE:** Recommended Default: Taxi overstay cap is 30 minutes unless guard extends on approval.

CAT-04: Vendor / Supplier

Purpose: Regular or one-time vendor supplying goods to departments or canteens.

Selfie: Mandatory

Field Name	Type	Required	Notes
Vendor Name (Person)	Text	Yes	
Vendor Phone	Phone	Yes	
Company / Firm Name	Text	Yes	
Goods / Service Description	Text	Yes	What is being supplied
Department / Contact Person	Text	Yes	Who authorized the vendor
Vehicle Number	Text	No	
Expected Exit Time	Time	Yes	
Selfie	Image	Yes	

CAT-05: Contractor / Construction Worker

Purpose: On-site construction or maintenance contractor and workers.

Selfie: Mandatory

Field Name	Type	Required	Notes
Contractor Lead Name	Text	Yes	Person in charge
Contractor Company	Text	Yes	
Phone	Phone	Yes	
Work Order / Project Reference	Text	No	If available
Number of Workers	Number	Yes	
Work Location on Campus	Text	Yes	Building/area name
Vehicle Number(s)	Text	No	Multiple vehicles separated by comma
Expected Exit Time	Time	Yes	
Selfie (Lead)	Image	Yes	Lead contractor selfie; workers listed by name
Duration of Project (days)	Number	No	For multi-day access planning

NOTE: Flag: Multi-day contractors should be issued a Recurring Pass by Admin to avoid daily re-registration. Design recurring pass feature in Phase 2.

CAT-06: Maintenance / Repair Service

Purpose: Plumbers, electricians, AC service, IT support, etc.

Selfie: Mandatory

Field Name	Type	Required	Notes
Technician Name	Text	Yes	
Phone	Phone	Yes	
Service Company	Text	Yes	
Service Type	Text	Yes	E.g., AC servicing, plumbing, elevator
Contact Person at University	Text	Yes	Who raised the service request
Work Location	Text	Yes	
Vehicle Number	Text	No	
Expected Exit Time	Time	Yes	
Selfie	Image	Yes	

CAT-07: Event Guest

Purpose: Guests attending a university event (seminar, convocation, sports, etc.).

Selfie: Mandatory

Field Name	Type	Required	Notes
Guest Name	Text	Yes	
Phone	Phone	Yes	
Event Name	Text	Yes	
Invited By / Host Name	Text	Yes	Faculty or department
Designation	Text	No	Industry, government, academic
Vehicle Number	Text	No	
Expected Exit Time	Time	Yes	
Selfie	Image	Yes	

 **NOTE:** Note: During large events, Admin can activate Event Mode which enables batch pre-registration by event organizers.

CAT-08: Emergency Service

Purpose: Ambulance, fire, police, emergency repair crews.

Selfie: No (skip selfie in emergency)

Field Name	Type	Required	Notes
Service Type	Dropdown	Yes	Ambulance / Fire / Police / Utility Emergency
Contact Name (Officer/Driver)	Text	Yes	
Contact Phone	Phone	Yes	
Vehicle Number	Text	Yes	
Responding To	Text	Yes	Brief description of emergency
Authorising Contact	Text	No	University contact who called

 **NOTE:** Emergency entries are auto-fast-tracked. Guard still logs the entry but can approve in one tap. Full logging maintained.

CAT-09: Interview Candidate

Purpose: Job applicants visiting for faculty/staff recruitment.

Selfie: Mandatory

Field Name	Type	Required	Notes
Candidate Name	Text	Yes	
Phone	Phone	Yes	
Interview For (Dept / Role)	Text	Yes	
Interview Contact Person	Text	Yes	HR or dept name

Interview Time	Time	Yes	Scheduled time
Vehicle Number	Text	No	
Selfie	Image	Yes	

CAT-10: Courier / Postal Service

Purpose: Postal department, BlueDart, Delhivery, FedEx type courier services.

Selfie: Optional

Field Name	Type	Required	Notes
Courier Company	Text	Yes	
Courier Person Name	Text	Yes	
Phone	Phone	Yes	
Consignment Number	Text	No	
Recipient Name / Department	Text	Yes	
Vehicle Number	Text	No	
Expected Exit Time	Time	Yes	Default: 30 min

CAT-11: Alumni Visitor

Purpose: Former students visiting the campus.

Selfie: Mandatory

Field Name	Type	Required	Notes
Alumni Name	Text	Yes	
Phone	Phone	Yes	
Roll Number / Batch Year	Text	Yes	Self-declared; no DB validation in Phase 1
Department / Programme	Text	Yes	
Purpose of Visit	Text	Yes	
Person to Meet	Text	No	Faculty or admin contact
Vehicle Number	Text	No	
Expected Exit Time	Time	Yes	
Selfie	Image	Yes	

CAT-12: Other / General Services

Purpose: Catch-all for any visitor not fitting above categories.

Selfie: Mandatory

Field Name	Type	Required	Notes
Visitor Name	Text	Yes	
Phone	Phone	Yes	
Purpose of Visit	Text	Yes	Mandatory free text
Person to Meet	Text	Yes	
Person's Department / Location	Text	No	
Vehicle Number	Text	No	
Expected Exit Time	Time	Yes	
Selfie	Image	Yes	

SECTION 13 — DETAILED END-TO-END USER FLOWS

Flow A: Standard Visitor Entry (All Categories)

18. Visitor arrives at gate. Scans QR code (physical poster at gate).
19. Mobile browser opens visitor form at: <https://vms.thapar.ac.in/visit?gate=FRONT>
20. Gate name is pre-filled. Visitor selects category.
21. Category-specific form loads. Visitor fills required fields.
22. Visitor captures selfie (live camera) or uploads from gallery.
23. Visitor submits form. System validates all required fields.
24. On success: Submission Reference ID shown (e.g., TU-2025-00342). Visit status = 'submitted'.
25. Guard's queue panel shows new entry with notification alert.
26. Guard opens the request, reviews details and selfie.
27. Guard compares selfie to person at gate (visual check).
28. Guard takes one of: Approve → Reject (reason) → Escalate (reason).
29. If Approved: Entry record created, status = 'entered'. Gate log updated.
30. Visitor is allowed in. Guard's screen shows 'Entry Logged — TU-2025-00342'.

Flow B: Escalation Flow

31. Guard clicks Escalate with a reason (e.g., 'Visitor refuses to show ID').
32. Visit status changes to 'escalated'. Removed from guard queue.
33. Supervisor sees escalated item highlighted in their multi-gate view.
34. Supervisor reviews full details, can call guard for clarification.
35. Supervisor takes one of: Approve / Reject / Hold / Return to Guard.
36. Hold status means visitor waits. Guard is notified.
37. All supervisor actions are logged.

Flow C: Exit Flow — Same Gate

38. Guard on duty at exit gate opens Exit Search panel.
39. Searches by name, vehicle number, or reference ID.
40. Active-inside records matching the search are shown.
41. Guard selects correct record. Visually verifies if needed.
42. Guard clicks Mark Exit. Status changes to 'exited'.
43. Exit gate, guard ID, and timestamp are logged.

Flow D: Exit Flow — Different Gate

44. Same as Flow C. The exit gate guard will see entry was at a different gate — this is displayed as informational, not an error.
45. Cross-gate exit is allowed and logged. Flagged in analytics.

Flow E: Emergency Visitor Entry

46. Guard (or visitor) selects Emergency Service category.
47. Minimal fields required. Guard can approve in single tap.
48. Emergency entry is logged with 'emergency' flag in audit.
49. Supervisor is auto-notified of emergency entry.
50. Exit is logged normally.

Flow F: Blacklisted Visitor Attempt

51. Visitor submits form. System checks phone number against blacklist.
52. Queue item appears with red WARNING: BLACKLISTED banner.
53. Guard sees blacklist reason from previous record.
54. Guard must make explicit decision: proceed to reject (standard) or escalate to Supervisor.
55. Auto-reject is NOT triggered — guard must act intentionally.

SECTION 14 — GATE-WISE QUEUE WORKFLOW

Each gate has an isolated, real-time queue:

Gate	QR Code	Queue Visibility
Front Gate	gate=FRONT	Guard(s) on Front Gate duty only + Supervisor + Admin
Western Gate	gate=WESTERN	Guard(s) on Western Gate duty only + Supervisor + Admin
Polytechnical Gate	gate=POLY	Guard(s) on Polytechnical Gate duty only + Supervisor + Admin
TSLAS Gate	gate=TSLAS	Guard(s) on TSLAS Gate duty only + Supervisor + Admin

Queue Item States

- pending — Submitted, awaiting guard action.
- under_review — Guard opened the record; clock starts.
- approved — Guard approved entry.
- rejected — Guard rejected with reason.
- escalated — Forwarded to Supervisor.
- on_hold — Supervisor placed on hold.

Queue Prioritisation Rules

- Emergency Service entries are pinned to top of queue.
- Escalated returns (Supervisor returns to Guard) appear highlighted.
- Within same priority, FIFO ordering applies.
- Items pending > 10 minutes turn amber. Items pending > 20 minutes turn red.

SECTION 15 — ENTRY WORKFLOW (DETAILED)

Pre-Conditions

- Visit record exists in 'submitted' or 'under_review' status.
- Guard is logged in and assigned to the correct gate.

Entry Action Sequence

56. Guard opens queue item → status moves to 'under_review'.
57. Guard reviews: name, category, selfie, key fields.
58. Guard performs visual identity check with person at gate.
59. Guard optionally calls student (for Parent visits) using student phone field.
60. Guard takes action: Approve / Reject / Escalate.
61. Approve: system writes entry_log record. Status → 'entered'.
62. Reject: system requires reason selection + optional free text. Status → 'rejected'.
63. Escalate: system requires reason. Status → 'escalated'. Supervisor notified.

Entry Log Record

- visit_id, guard_id, guard_name, gate_id, gate_name, action='entry_approved', timestamp, guard_ip
-  **NOTE:** Guards cannot edit an entry record after writing. Only Head Admin can void with full audit.

SECTION 16 — EXIT WORKFLOW (DETAILED)

Exit Search Panel

- Guard opens Exit Search (dedicated tab/panel separate from entry queue).
- Default view: all active-inside visitors for all gates (searchable).
- Search fields: Name (fuzzy), Phone, Vehicle Number, Reference ID, Category.
- Results show: Name, Category, Entry Gate, Entry Time, Time Inside.

Exit Marking

- Guard locates visitor record.
- Guard optionally expands to verify details.
- Guard clicks Mark Exit.
- System writes exit_log record with exit_gate_id, exit_guard_id, exit_timestamp.
- Status changes to 'exited'.
- Record moves to today's exited list; no longer appears in active-inside.

Unknown Exit (Edge Case)

- If guard needs to log an exit with no matching entry record: 'Manual Exit — Unverified Entry' option with mandatory explanation. Logged as anomaly.

SECTION 17 — EXCEPTION & EDGE CASE HANDLING

Edge Case	Recommended Handling
Visitor refreshes form and submits twice	Deduplication window: same phone + same gate + 10 min → show existing status
Visitor exit before entry logged	Not allowed via normal flow; manual override with supervisor approval + audit entry
Guard logs out mid-queue	Queue items remain in 'submitted' state and are reassigned to next guard logging into that gate
Two guards on same gate concurrently	Both see same queue; first guard to open an item claims it (optimistic locking)
Visitor has no phone for dedup	Use name + gate + 5-min window as secondary deduplication signal
Selfie upload fails (network issue)	Form stores selfie in browser memory; retry on reconnect. If camera unavailable, upload fallback shown
Expected exit time not provided	System assigns category default (configurable by Admin). Overstay timer starts from default
Vehicle number absent for taxi	Guard can proceed but system shows yellow flag 'vehicle number not provided'
Visitor tries to re-enter same day	System shows prior visit history. Guard decides. Duplicate active session blocked — previous must be exited first
Supervisor makes bulk approval	Not recommended. Each entry must be reviewed individually. No bulk approve feature in Phase 1

Network outage during submission	Visitor's form submission is queued client-side (service worker). Retried when connection restores
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SECTION 18 — RECOMMENDED APPROVAL RULES

- All approvals are manual. No auto-approve in Phase 1.
 - Guard can Quick Approve returning visitors (previous visit history shown).
 - Selfie review is mandatory before any approval; system shows selfie prominently in detail view.
 - Categories with mandatory selfie: rejection is allowed if selfie is absent without valid reason.
 - A rejected entry cannot be resubmitted by the visitor without approval from Supervisor.
 - An escalated entry bypasses the guard queue and is handled only by Supervisor.
 - Once 'entered', status cannot be changed back to pending/under_review.
 - A visit can be cancelled (status = 'cancelled') only by Supervisor or Admin, with reason.
- ✓ **RECOMMENDED DEFAULT:** Edit is allowed on submitted/pending records by the system (e.g., if guard edits a typo). Once approved/entered, edits require override with audit log.

SECTION 19 — SEARCH & RETRIEVAL RULES

Active-Inside Search (Exit Guard)

- Default: Shows all active-inside visitors across all gates.
- Guard can filter by their current gate for convenience.
- Search fields: name (contains), phone (exact or last 4 digits), vehicle number (contains), reference ID (exact), category dropdown, company name.
- Results sorted by time inside (longest first) to surface overstay risks.

General Search (Admin / Supervisor)

- Search all visits by date range, gate, status, category, guard who processed.
- Search by visitor name, phone, vehicle, student name, company.
- Supervisor export of search results to CSV.

SECTION 20 — GUARD SHIFT / GATE ASSIGNMENT LOGIC

Shift Assignment at Login

- After login, guard is prompted: 'Which gate are you currently posted at?' (dropdown of 4 gates).
- Guard's session is tagged with the selected gate.
- Guard sees only that gate's queue.
- Guard can change gate (e.g., during shift handover) from profile menu — this creates a new gate_assignment log entry.

Shift Accountability

- Every entry/exit action in a guard's session is associated with their gate assignment at that moment.
- Shift summary: total approved, rejected, escalated, and exits processed by guard per shift.
- Supervisor can view guard shift activity timeline.

Concurrent Guard Logic

- Multiple guards can be logged into the same gate (e.g., shift handover overlap).
- Queue item claimed by the first guard to open it. Others see it as 'being reviewed'.
- If claiming guard takes no action within 5 minutes, item reverts to unclaimed.

SECTION 21 — FAST APPROVAL / FAST EXIT FEATURES

Quick Approve

- Available when visitor has at least 1 prior approved visit on record.
- Prior visit history shown inline: last visit date, gate, category.
- Guard can click Quick Approve — still logs full entry record.
- Quick Approve is not available for: Emergency (always full log), Blacklisted visitors (always full review).

Fast Exit

- Dedicated exit panel is always one tab away in guard interface.
- Recent exits shown in Quick Recent list (last 20 exits from any gate).
- Guard can search by vehicle number from gate (for drive-through exits).
- Mark Exit is a single-tap action with confirmation dialog.

Quick Reject Shortcuts

- Preset reject reasons: 'Incomplete ID', 'Selfie mismatch', 'Blacklisted', 'Visit time not allowed', 'No matching student record', 'Vehicle not permitted', 'Other (specify)'.
- Guard selects preset reason → confirms → rejected with reason recorded.

SECTION 22 — DASHBOARD REQUIREMENTS

Guard Dashboard (Single Gate)

- Live queue count with status breakdown (pending / under_review / escalated).
- Active inside count for current gate.
- Overstay alerts for current gate.
- Today's stats: approved, rejected, exited.
- My shift stats: actions taken since login.

Supervisor Dashboard (All Gates)

- Gate-by-gate live queue counts in a card/tile layout.
- All-gate active-inside total count.
- Escalated items list (cross-gate).
- Overstay alerts across all gates.
- Guard activity today (per guard: # approved, rejected, escalated).
- Category distribution chart for today.

Head Admin Dashboard (Full System)

- All supervisor views plus:
- System health indicators (API uptime, queue processing time avg).
- Audit trail summary (recent critical events).
- User management quick-access.
- Weekly/monthly trend charts.
- Blacklist count and recent additions.
- Gate load comparison chart.

SECTION 23 — REPORTS & CSV EXPORT REQUIREMENTS

Report Name	Filters Available	Access Level
Daily Visit Log	Gate, Date, Category, Status	Supervisor (own gate), Admin (all)
Active Inside Visitors	Gate, Category, Time inside > N hours	Guard, Supervisor, Admin
Overstay Report	Date range, Gate, Category	Supervisor, Admin
Guard Performance Report	Guard name, Gate, Date range	Supervisor, Admin
Rejected Entries Report	Gate, Reason, Date	Supervisor, Admin
Gate Load Report	Date range, Hourly view	Supervisor, Admin
Blacklist Report	Status, Date added	Admin only
Full Audit Trail Export	Date range, Event type, User	Admin only

CSV Export Specifications

- All reports are exportable as CSV with UTF-8 encoding.
- Export includes all visible columns in the filtered view.

- Selfie images are NOT included in CSV; a photo_reference_id column links to secure photo storage.
- Large exports (> 10,000 rows) are queued and emailed as secure download links.

SECTION 24 — ALERTING & NOTIFICATION REQUIREMENTS

Alert Channels

- In-app notification banner (primary — always active).
- Sound/vibration for new queue item (guard panel, configurable).
- Email alert to supervisor for critical overdue alerts.
- Optional SMS integration for critical alerts (Phase 2).

Alert Types

Alert Type	Trigger	Recipient	Priority
New Visitor in Queue	Form submitted at guard's gate	On-duty guard	Normal
Queue Item Pending > 10 min	Item age exceeds 10 minutes	Guard + Supervisor	Medium
Overstay Reminder	30 min before expected exit	On-duty guard	Normal
Overstay Escalation	At or past expected exit time	Guard + Supervisor	High
Critical Overstay	+4 hours past expected exit	Supervisor + Admin	Critical
Blacklist Hit	Blacklisted visitor submits form	Guard + Supervisor	High
Escalation Received	Guard escalates a visit	Supervisor	High
Emergency Entry	Emergency category approved	Supervisor + Admin	Critical
Concurrent Login Alert	Guard account logged in from 2+ IPs	Admin	High

SECTION 25 — AUTHENTICATION & AUTHORIZATION DESIGN

Authentication Model

- JWT (JSON Web Token) based authentication for all internal users (guards, supervisors, admin).
- Visitor form is public — no authentication required. Rate limiting applied per IP.
- Access tokens: 4-hour expiry. Refresh tokens: 24-hour expiry (single use).
- Guard login flow: Username + Password → Gate Selection → Session created.
- Admin/Supervisor login: Username + Password + TOTP (2FA, Phase 2 for Supervisor, mandatory for Admin).

Password Policy

- Minimum 10 characters. Must include uppercase, lowercase, number, special character.
- Passwords hashed with bcrypt (cost factor 12). Never stored in plaintext.
- Password reset via admin-generated reset link (no self-service email in Phase 1).
- Force password change on first login.

Session Rules

- Guard sessions expire after 4 hours of inactivity.
 - Concurrent sessions from different IPs trigger admin alert (not auto-block in Phase 1).
 - Logout invalidates refresh token server-side.
 - Session gate assignment is immutable after login (re-login required to change gate).
- ✓ **RECOMMENDED DEFAULT:** Guard re-login required to switch gate assignment. This ensures gate logs are clean.

Failed Login Handling

- After 5 failed attempts: account locked for 30 minutes.
- Failed attempts are logged with timestamp and IP.
- Admin receives alert if 10+ failed attempts in 5 minutes (brute force signal).
- Admin can manually unlock accounts.

Role-Based Access Control (RBAC)

- Three roles: guard, supervisor, head_admin.
- Permissions are enforced on both API (middleware) and UI (component visibility).
- No permission inheritance exceptions — roles are discrete.
- API endpoints return 403 Forbidden for unauthorized role attempts; all 403s are logged.

Impersonation Prevention

- No admin impersonation feature. Admin cannot log in as another user.
- All audit events record the actual logged-in user, not an assumed identity.

SECTION 26 — AUDIT TRAIL DESIGN

Principles

- All audit events are append-only writes to a separate audit log database or schema.
- No update or delete operations are permitted on audit records, even by Admin.
- Audit logs are cryptographically chained (each record includes hash of previous record) in Phase 2.

Recommended Audit Events

Event Code	Trigger	Data Captured
VISIT_SUBMITTED	Visitor submits form	
VISIT_REVIEWED	Guard opens a queue item	
VISIT_APPROVED	Guard approves entry	
VISIT_REJECTED	Guard rejects entry	
VISIT_ESCALATED	Guard escalates	
VISIT_ENTERED	Entry logged	
VISIT_EXITED	Exit logged	
VISIT_CANCELLED	Supervisor/Admin cancels	
VISIT_OVERRIDDEN	Admin voids/overrides record	
ALERT_FIRED	Overstay alert triggered	
BLACKLIST_ADDED	Visitor added to blacklist	
BLACKLIST_HIT	Blacklisted visitor attempted entry	
USER_LOGIN	Any user logs in	
USER_LOGOUT	User logs out	
AUTH_FAILED	Failed login attempt	
GUARD_GATE_CHANGE	Guard changes gate assignment	
ADMIN_ACTION	Any admin user management action	

SECTION 27 — DATA MODEL / DATABASE SCHEMA

NOTE: All tables include: `created_at TIMESTAMPTZ NOT NULL DEFAULT now()`, `updated_at TIMESTAMPTZ`. Soft deletes via `is_deleted BOOLEAN` where applicable. UUIDs used for all primary keys.

Table: users

Stores all internal system users (guards, supervisors, admin).

Column	Type	Constraints	Description
id	UUID	PK	Auto-generated UUID
username	VARCHAR(50)	UNIQUE NOT NULL	Login username
password_hash	VARCHAR(255)	NOT NULL	bcrypt hash
full_name	VARCHAR(100)	NOT NULL	Display name
role	ENUM	NOT NULL	guard supervisor head_admin
phone	VARCHAR(15)	UNIQUE	Contact for alerts
email	VARCHAR(100)	UNIQUE	Admin/Supervisor email for reports
is_active	BOOLEAN	DEFAULT TRUE	Account enabled flag
last_login_at	TIMESTAMPTZ		Last successful login
failed_login_count	INTEGER	DEFAULT 0	Reset on success
locked_until	TIMESTAMPTZ	NULLABLE	Lockout expiry
force_password_change	BOOLEAN	DEFAULT TRUE	Force change on first login

Table: gates

Reference table for 4 university gates.

Column	Type	Constraints	Description
id	UUID	PK	
name	VARCHAR(100)	UNIQUE NOT NULL	e.g., Front Gate
code	VARCHAR(20)	UNIQUE NOT NULL	e.g., FRONT, WESTERN, POLY, TSLAS
qr_token	VARCHAR(255)	UNIQUE NOT NULL	Embedded in QR code URL
is_active	BOOLEAN	DEFAULT TRUE	Can disable a gate

Table: guard_gate_assignments

Records which gate a guard is on duty at per session.

Column	Type	Constraints	Description
id	UUID	PK	

guard_id	UUID	FK → users	
gate_id	UUID	FK → gates	
assigned_at	TIMESTAMPZ	NOT NULL	When guard selected gate on login
released_at	TIMESTAMPZ	NULLABLE	When guard logged out or changed gate
session_id	VARCHAR(255)	NOT NULL	Links to auth session

Table: visitor_categories

Admin-configurable visitor categories.

Column	Type	Constraints	Description
id	UUID	PK	
code	VARCHAR(30)	UNIQUE NOT NULL	e.g., PARENT, DELIVERY, TAXI
display_name	VARCHAR(100)	NOT NULL	Shown on form
selfie_required	BOOLEAN	DEFAULT TRUE	
default_max_duration_minutes	INTEGER	NOT NULL	Overstay cap; default varies by category
is_active	BOOLEAN	DEFAULT TRUE	Admin can disable categories
sort_order	INTEGER		Display order on form

Table: visits

Core visit record — one per visitor visit.

Column	Type	Constraints	Description
id	UUID	PK	
reference_id	VARCHAR(20)	UNIQUE NOT NULL	Human-readable: TU-2025-00342
gate_id	UUID	FK → gates	Gate where QR was scanned
category_id	UUID	FK → visitor_categories	
visitor_name	VARCHAR(150)	NOT NULL	
visitor_phone	VARCHAR(15)	NOT NULL	
visitor_id_type	VARCHAR(30)	NULLABLE	Aadhaar / DL etc.
visitor_id_last4	VARCHAR(4)	NULLABLE	Last 4 digits only
selfie_object_key	VARCHAR(500)	NULLABLE	S3/storage path to compressed image
selfie_captured_at	TIMESTAMPZ	NULLABLE	

selfie_hash	VARCHAR(64)	NULLABLE	SHA-256; replaces image after retention period
form_data	JSONB	NOT NULL	All category-specific fields as JSON
expected_exit_at	TIMESTAMPTZ	NULLABLE	From form or category default
status	ENUM	NOT NULL	See status lifecycle below
is_blacklist_hit	BOOLEAN	DEFAULT FALSE	Flagged at submission time
submitted_at	TIMESTAMPTZ	NOT NULL	Form submission time
reviewed_by	UUID	FK → users NULLABLE	Guard who claimed the item
reviewed_at	TIMESTAMPTZ	NULLABLE	
decided_by	UUID	FK → users NULLABLE	Guard/supervisor who made final decision
decided_at	TIMESTAMPTZ	NULLABLE	
reject_reason_code	VARCHAR(50)	NULLABLE	Preset reason code
reject_reason_text	TEXT	NULLABLE	Free text
escalated_by	UUID	FK → users NULLABLE	
escalated_at	TIMESTAMPTZ	NULLABLE	
escalation_reason	TEXT	NULLABLE	
entry_guard_id	UUID	FK → users NULLABLE	
entry_gate_id	UUID	FK → gates NULLABLE	May differ from initial gate_id in edge cases
entry_at	TIMESTAMPTZ	NULLABLE	Actual physical entry time
exit_guard_id	UUID	FK → users NULLABLE	
exit_gate_id	UUID	FK → gates NULLABLE	May differ from entry gate
exit_at	TIMESTAMPTZ	NULLABLE	
duration_minutes	INTEGER	NULLABLE	Computed on exit
is_emergency	BOOLEAN	DEFAULT FALSE	Emergency category flag
vehicle_number	VARCHAR(20)	NULLABLE	Indexed for exit search

Table: visit_status_lifecycle

Reference: valid status transitions.

Column	Type	Constraints	Description
submitted	→ under_review	Guard opens	Initial state after form submission
under_review	→ approved / rejected / escalated	Guard action	Guard claimed item
approved	→ entered	Entry logged	Guard confirms person has passed gate
entered	→ exited / overstayed	Exit logged or time exceeded	Active inside state
overstayed	→ exited	Exit eventually logged	Alert state, visit continues
escalated	→ approved / rejected / on_hold / back_to_guard	Supervisor	
on_hold	→ approved / rejected / cancelled	Supervisor	
rejected	Terminal	—	Cannot be reactivated without Supervisor
exited	Terminal	—	Visit complete
cancelled	Terminal	—	Supervisor/Admin only
expired	Terminal	—	Submitted but no guard action within 24h

Table: blacklist

Persons barred or flagged from campus.

Column	Type	Constraints	Description
id	UUID	PK	
visitor_phone	VARCHAR(15)	INDEXED	Primary match key
visitor_name	VARCHAR(150)		Supplementary
reason	TEXT	NOT NULL	Why blacklisted
status	ENUM	NOT NULL	flagged blacklisted cleared
added_by	UUID	FK → users	
reviewed_by	UUID	FK → users NULLABLE	Admin confirmation
visit_id_reference	UUID	FK → visits NULLABLE	Visit that triggered blacklist
expires_at	TIMESTAMPZ	NULLABLE	Optional expiry; NULL = permanent

Table: audit_logs

Immutable audit trail — append-only.

Column	Type	Constraints	Description
id	UUID	PK	
event_code	VARCHAR(50)	NOT NULL INDEXED	e.g., VISIT_APPROVED
actor_id	UUID	FK → users NULLABLE	NULL if visitor-side action
actor_role	VARCHAR(30)	NULLABLE	Role at time of action
visit_id	UUID	FK → visits NULLABLE	Related visit
gate_id	UUID	FK → gates NULLABLE	Gate context
event_data	JSONB	NOT NULL	Full event payload
ip_address	INET	NULLABLE	Actor IP
occurred_at	TIMESTAMPTZ	NOT NULL DEFAULT now()	Immutable

Table: overstay_alerts

Tracks alert state per visit.

Column	Type	Constraints	Description
id	UUID	PK	
visit_id	UUID	FK → visits NOT NULL	
alert_tier	ENUM	NOT NULL	reminder escalation critical
fired_at	TIMESTAMPTZ	NOT NULL	
acknowledged_by	UUID	FK → users NULLABLE	
acknowledged_at	TIMESTAMPTZ	NULLABLE	
is_resolved	BOOLEAN	DEFAULT FALSE	Auto-resolved on exit

SECTION 28 — SUGGESTED API ENDPOINTS

NOTE: All authenticated endpoints require Authorization: Bearer <token> header. All POST/PUT bodies are JSON. All responses are JSON. API version prefix: /api/v1/ in production.

Method	Endpoint	Auth	Description
POST	/api/visits/submit	Public (rate-limited)	Submit visitor form
GET	/api/visits/status/:referenceId	Public	Get visit status by reference ID
GET	/api/gates/:gateCode/qr-info	Public	Get gate display info from QR token
GET	/api/categories	Public	List active visitor categories + fields
GET	/api/guard/queue	Guard+	Get pending queue for guard's current gate
GET	/api/guard/queue/:visitId	Guard+	Get full visit details
POST	/api/guard/queue/:visitId/approve	Guard+	Approve entry (logs entry record)
POST	/api/guard/queue/:visitId/reject	Guard+	Reject with reason code + text
POST	/api/guard/queue/:visitId/escalate	Guard+	Escalate to supervisor with reason
GET	/api/guard/active-inside	Guard+	Search active-inside visitors
POST	/api/guard/exit/:visitId	Guard+	Mark exit for a visitor
GET	/api/guard/my-shift-stats	Guard+	Today's stats for logged-in guard
GET	/api/supervisor/all-queues	Supervisor+	All gate queues with counts
GET	/api/supervisor/escalations	Supervisor+	All pending escalations
POST	/api/supervisor/escalations/:id/resolve	Supervisor+	Approve/reject/hold escalated visit
POST	/api/supervisor/visits/:id/cancel	Supervisor+	Cancel a visit with reason
POST	/api/supervisor/visits/:id/override	Supervisor+	Override decision with reason + audit
GET	/api/supervisor/reports	Supervisor+	Get reports (own gate for sup; all for admin)
POST	/api/supervisor/export/csv	Supervisor+	Request CSV export
POST	/api/supervisor/watchlist	Supervisor+	Add visitor to watchlist
GET	/api/admin/users	Admin	List all users
POST	/api/admin/users	Admin	Create user (guard/supervisor)
PUT	/api/admin/users/:id	Admin	Update user details or role
POST	/api/admin/users/:id/deactivate	Admin	Deactivate account
POST	/api/admin/users/:id/reset-password	Admin	Generate password reset token
GET	/api/admin/audit-trail	Admin	Full audit trail with filters
GET	/api/admin/blacklist	Admin	View blacklist

POST	/api/admin/blacklist	Admin	Add to blacklist
DELETE	/api/admin/blacklist/:id	Admin	Remove from blacklist
PUT	/api/admin/categories/:id	Admin	Configure category fields/thresholds
GET	/api/admin/dashboard	Admin	Full system dashboard data
POST	/api/admin/emergency-mode	Supervisor+	Toggle emergency mode for a gate
POST	/api/auth/login	Public	Login, returns JWT + refresh token
POST	/api/auth/logout	Authenticated	Invalidate session
POST	/api/auth/refresh	Authenticated	Refresh access token
POST	/api/auth/change-password	Authenticated	Change own password

SECTION 29 — SCREEN-BY-SCREEN UI MODULE BREAKDOWN

Visitor-Side (Public Web)

SCR-V1: Gate Landing Page

- Gate name displayed prominently. University branding.
- Category selector as icon-grid (Parent, Delivery, Taxi, etc.).
- Language selector (Phase 2). Privacy notice link.

SCR-V2: Visitor Form

- Dynamic fields based on selected category.
- Selfie capture button → opens device camera. Preview before submit.
- Progress indicator for required fields.
- Submit button disabled until all required fields valid.
- Estimated wait time displayed (based on current queue depth).

SCR-V3: Submission Confirmation

- Reference ID displayed in large font. Copy button.
- 'Please show this screen to the guard' instruction.
- Current status display (updates via polling every 5 seconds).
- Guard contact number (optional, admin-configurable).

Guard Panel

SCR-G1: Guard Login

- Username / Password fields. Show/hide password toggle.
- After credentials: Gate Selection dropdown (4 gates). Confirm button.
- Last login info displayed after success.

SCR-G2: Guard Main Queue

- Live queue list with cards. Each card: name, category icon, time submitted, selfie thumbnail.
- Color coding: white (new) → amber (>10min) → red (>20min).
- Tab bar: Queue | Active Inside | Exit | My Stats.
- Notification badge on Queue tab.

SCR-G3: Visit Detail View

- Selfie displayed prominently (large, top of screen).
- All form fields displayed in a clean list.
- Previous visits section (if any) showing last 3 visits.
- Blacklist/watchlist warning banner (if applicable).
- Action buttons: Approve (green) | Reject (red) | Escalate (orange).
- Quick Approve highlighted if returning visitor.

SCR-G4: Reject Modal

- Reason selector (preset codes). Optional free-text.
- Confirm button.

SCR-G5: Exit Search

- Search bar (prominent). Filter chips: All Gates / Current Gate, Category.
- Results list: name, category, entry time, entry gate, time inside.
- Mark Exit button per result (single tap + confirm).

SCR-G6: My Shift Stats

- Today: Approved / Rejected / Escalated / Exits processed.
- Current shift since login.
- Recent activity timeline.

Supervisor Panel

SCR-S1: Multi-Gate Overview

- 4 gate cards with live queue counts and alert indicators.
- Escalated items count (prominent).

SCR-S2: Escalation Queue

- All pending escalations. Filter by gate.
- Same detail view as guard SCR-G3 with additional context (escalation reason).
- Actions: Approve / Reject / Hold / Return to Guard.

SCR-S3: All-Gate Queue View

- Combined queue across gates. Gate filter. Status filter.

SCR-S4: Reports

- Report type selector. Date range, gate filter. Preview table. Export CSV button.

Admin Panel

SCR-A1: Full Dashboard

- System health widget. All gate status. Alert summaries. Recent audit events. Charts.

SCR-A2: User Management

- Table of all users. Create / Edit / Deactivate buttons. Role filter.

SCR-A3: Audit Trail

- Filterable event log. Date range, event type, user, gate filters. CSV export.

SCR-A4: Blacklist Management

- Current blacklist. Add entry form. Remove with audit trail.

SCR-A5: System Settings

- Category configuration. Overstay thresholds. Gate management. Alert settings.

SECTION 30 — ADMIN CONTROL FEATURES

- Create, edit, deactivate guard and supervisor accounts.
- Assign supervisors to specific gates or all gates.
- Configure visitor categories: fields, selfie requirement, overstay defaults.
- Manage blacklist: add, review, remove.
- Full audit trail access with export.
- Override any visit record with mandatory reason and audit entry.
- Configure system-wide settings: session timeout, alert thresholds, QR tokens.
- Emergency mode toggle for any gate.
- Photo retention management: extend or manually trigger deletion.

SECTION 31 — SUPERVISOR FEATURES

- View all-gate queues and active-inside visitors.
- Handle escalations: approve, reject, hold, return to guard.
- Override guard decisions on individual visits with mandatory reason.
- Cancel visits with reason.
- Generate and export reports for gates in their scope.
- Add visitors to watchlist (pending Admin confirmation for blacklist).
- View guard performance on their assigned gates.
- Trigger emergency mode (Supervisor+ permission).

SECTION 32 — GUARD FEATURES

- Login and gate selection.
- View and process live entry queue for their gate.
- Approve, reject, or escalate visitor entries.
- View visitor selfie and prior visit history.
- Mark exit for any active-inside visitor across all gates.
- View own shift statistics.
- Fast exit search by vehicle number or name.

SECTION 33 — OPERATIONAL POLICIES

Entry Policy

- No visitor enters without a logged approved record.
- Emergency services are logged immediately and may enter concurrently with logging.
- Guard approval is the final human gate. System does not auto-approve.

Exit Policy

- Any guard at any gate can mark an exit.
- Exit does not require the same guard who approved entry.
- Visitors are assumed still inside if no exit is logged. Overstay alerts apply.

Re-entry Policy

✔ **RECOMMENDED DEFAULT:** A visitor may not have two simultaneous active visits. Previous visit must be exited before re-entry is permitted.

- Re-entry on same day is allowed but flagged in guard's view for awareness.

Recurring Visitor Policy

- Phase 1: Recurring visitors register each time via QR form. System shows prior visit history to guard.
- Phase 2: Admin-issued Recurring Pass with pre-approved access window for known contractors.

SECTION 34 — PRIVACY, SECURITY & DATA RETENTION ANALYSIS

Data Classification

Data Type	Sensitivity	Retention Policy	Access Control
Selfie images	HIGH	12 months active → SHA-256 hash only	Guard (view only), Admin (manage)
Visitor phone number	MEDIUM	Stored hashed in audit; plain in visits table (encrypted at rest)	All roles
Government ID number	HIGH	Last 4 digits stored only. Full number shown to guard live but not persisted	Guard (view live only)
Audit logs	HIGH	Permanent (append-only)	Admin only
Visit records (without photos)	MEDIUM	5 years (archival after 1 year)	All roles (per RBAC)
Guard login sessions	LOW	6 months	Admin only

Compliance Considerations

- India's Digital Personal Data Protection (DPDP) Act 2023 applies. Consent must be obtained at form submission.
- A privacy notice must be shown before form submission (brief, plain language).
- Visitors have the right to request data deletion under DPDP. Admin must be able to execute verified deletion requests within 30 days.
- Government ID full numbers must never be stored in plaintext. Display-only, then cleared from memory.

🚩 **RISK/FLAG:** Storing selfies forever without consent renewal or legal basis is a compliance violation under DPDP. The recommended 12-month retention policy with hashing is strongly advised.

SECTION 35 — ABUSE / FRAUD / MISUSE PREVENTION

- Rate limiting on visitor form: max 5 submissions per phone number per day per gate.
- Duplicate submission detection: same phone + same gate + 10-minute window.
- Guard action logs: any unusually high approval rate (> 50 approvals/hour) triggers supervisor alert.
- Selfie quality check: basic face detection (client-side) to prevent blank photo uploads.
- QR code token validation: gate QR tokens are unique and validated server-side. Fake QR codes produce invalid gate ID error.
- Session-based gate assignment: guard cannot approve entries from a gate they're not logged into.
- IP-based concurrent session detection for guard accounts.
- Admin actions (blacklist, override, delete) all require explicit confirmation dialogs plus mandatory reason fields.
- Audit trail integrity: periodic checksums verify no records have been tampered with.

SECTION 36 — SCALABILITY RECOMMENDATIONS

- Backend: Stateless REST API with horizontal scaling via container orchestration (Kubernetes or Docker Compose with Nginx load balancer).
- Database: PostgreSQL with read replicas for dashboard queries. Connection pooling via PgBouncer.
- Photo Storage: Object storage (AWS S3, Cloudflare R2, or MinIO for on-premise). Images never stored in DB.
- Queue Processing: Real-time queue via WebSocket (Socket.IO or native WS). Fallback to 10-second polling.
- Caching: Redis for session management, active-inside count cache, and overstay alert job state.
- Overstay Job: Background worker (Node.js cron or Celery) polling active visits every 5 minutes.
- CDN: Static assets (visitor form, guard panel) served via CDN for fast mobile load.

SECTION 37 — FUTURE ENHANCEMENTS & AI FEATURES

Phase 2 Features

- AI Face Match: Compare submission selfie to a second photo taken at gate. Flag mismatch for guard review.
- Recurring Pass System: Admin-issued time-bound passes for known contractors. Pass QR replaces form.
- Suspicious Pattern Detection: ML model flags anomalous visit patterns (same person multiple rejected entries, unusual hours).
- SMS/WhatsApp Notifications: Status updates to visitor's phone number.
- Biometric Integration: Optional fingerprint or face scan at gate barrier hardware.
- Student Verification Hook: When university student DB becomes API-accessible, auto-validate student name/hostel.
- Event Pre-Registration: Event organizers can pre-register guest lists; visitors check in with name-based lookup.
- Multi-language Form: Punjabi, Hindi form support via i18n framework.
- Guard Mobile Native App: Dedicated Android/iOS app for guards with push notifications and offline queue.
- Predictive Gate Load: ML model predicts visitor peaks based on university calendar, suggests guard staffing.
- Visitor ID OCR: Auto-fill form fields by photographing ID card.

SECTION 38 — RECOMMENDED TECH STACK

Layer	Technology	Rationale
Frontend (Visitor Form)	React + Vite + TailwindCSS	
Frontend (Guard / Admin)	React + Vite + TailwindCSS + shadcn/ui	
Backend API	Node.js + Express (or Fastify)	
Database	PostgreSQL 15+	
Cache / Sessions	Redis 7	
Photo Storage	AWS S3 / Cloudflare R2 / MinIO	
Real-time Queue	Socket.IO over WebSocket	
Auth	JWT (jsonwebtoken) + bcrypt	
Background Jobs	node-cron / BullMQ	

Deployment	Docker + Docker Compose (dev); Kubernetes (prod)
Reverse Proxy	Nginx
Monitoring	Prometheus + Grafana (or Datadog)
Error Tracking	Sentry

SECTION 39 — MVP vs PHASE-2 FEATURE SPLIT

MVP (Phase 1) — Deliverable in 8–12 Weeks

- Visitor form (all 12 categories) with dynamic fields, selfie capture, QR gate detection.
- Guard queue panel: view, approve, reject, escalate.
- Entry and exit logging with cross-gate search.
- Overstay alerts (in-app, basic three-tier).
- Supervisor multi-gate view and escalation handling.
- Head Admin dashboard, user management, basic audit trail.
- Blacklist (basic: add/view/remove).
- Daily visit report + CSV export.
- JWT authentication + RBAC.
- PostgreSQL schema + S3 photo storage.

Phase 2 — Post-Launch Enhancements

- AI face matching and selfie comparison.
- SMS/WhatsApp visitor notifications.
- Recurring pass system for contractors.
- Native guard mobile app with push notifications.
- Advanced analytics dashboard (ML-based predictions).
- Student database integration hook.
- Event pre-registration module.
- Cryptographically chained audit logs.
- 2FA for Admin and Supervisor accounts.
- Visitor ID OCR auto-fill.
- Multi-language support.

SECTION 40 — TESTING STRATEGY

Unit Tests

- Business logic: status transition guards, overstay calculation, deduplication logic, permission checks.
- Coverage target: 80%+ for core services.

Integration Tests

- API endpoint tests: all routes with valid/invalid auth, correct role enforcement.
- Database integration: CRUD operations, constraint validation.

End-to-End Tests

- Visitor form submission → Guard approval → Entry log → Exit → Report export.
- Escalation flow. Blacklist hit flow. Overstay alert firing.
- Cross-gate exit scenario.

Load Tests

- Simulate 100 simultaneous form submissions across 4 gates.
- Guard queue handling under 50 concurrent requests.
- Exit search under 500 active-inside records.

Security Tests

- OWASP Top 10 penetration testing checklist.
- Role escalation attempt (guard trying to access admin endpoints).
- SQL injection and XSS on all input fields.
- JWT forgery and replay attack tests.

SECTION 41 — DEPLOYMENT CONSIDERATIONS

- Domain: vms.thapar.ac.in (or subdomain). HTTPS via Let's Encrypt or institutional certificate.
- On-premise deployment recommended for Phase 1 to satisfy university data residency requirements.
- DB backups: automated daily to off-site S3 bucket. Encrypted at rest.
- Environment separation: Development → Staging → Production with separate DB instances.
- Initial data seeding: 4 gates, 12 categories, Head Admin account.
- QR code generation: Admin generates gate QR codes from the system. Printed and laminated at each gate.
- Guard onboarding: IT provisions accounts; guards complete first-login flow with forced password change.
- Go-live plan: 2-week parallel run with paper registers alongside the system. Full cutover after validation.

SECTION 42 — FINAL PRODUCT RECOMMENDATIONS

- Start with a focused MVP: nail the guard queue experience first. Speed and reliability at the gate is the #1 success metric.
- The visitor form must be under 3 screen-lengths on a mobile phone. Do not add unnecessary fields.
- Design the guard panel for one-handed mobile use. Approve/Reject must be easily reachable thumb targets.
- Build the audit trail from Day 1. Retrofitting it is significantly harder and creates liability gaps.
- Test with real guards at the gate before full launch. Gate conditions (sun glare, noise, rushed interactions) are different from office testing.
- Photo retention policy decision must be formalized and approved by university administration before launch.
- Publish a simple one-page visitor guide at each gate alongside the QR code.

SECTION 43 — DELIVERABLES SUMMARY

Deliverable H: Recommended Status Values

Status	Meaning	Who Can Set	Terminal?
submitted	Form submitted; in guard queue	System (auto)	No
under_review	Guard opened item; reviewing	Guard	No
escalated	Forwarded to Supervisor	Guard	No
on_hold	Supervisor placed on hold	Supervisor+	No
approved	Decision made; entry authorised	Guard/Supervisor/Admin	No
entered	Guard confirmed physical entry	Guard+	No
overstayed	Inside beyond expected exit time	System (auto)	No
exited	Exit logged	Guard+	Yes
rejected	Entry denied by guard	Guard/Supervisor/Admin	Yes
cancelled	Cancelled by Supervisor/Admin	Supervisor+	Yes
expired	No guard action within 24 hours	System (auto)	Yes

Deliverable F: Key Validations

- Phone number: 10-digit Indian mobile number format (regex validated).
- Vehicle number: Indian vehicle number plate format (e.g., PB 11 AA 1234).
- Selfie: File type must be image/jpeg or image/png. Min 200×200px. Max 5MB (compressed to 800KB).
- Expected exit time: Must be in future. Must be within 24 hours (configurable max).
- Government ID number: Masked to last 4 digits on storage. Full number validated against type-specific format.
- Gate QR token: Must match a valid gate record in DB.
- All text fields: Strip HTML/script tags (XSS prevention). Max length enforced per field.
- Number of workers (Contractor): Integer, min 1, max 100.
- Student year: Must be 1–5 (dropdown; not free text).
- Category selection: Must be a valid active category ID from the server-provided list.

Deliverable E: Edge Cases

- Visitor submits form after guard's shift ends and no guard is logged in: Form is queued. Next guard sees it on login.
- Guard network drops mid-approval: Approval request is idempotent (same visit_id + action = same result if retried).
- Two guards simultaneously try to approve same visitor: First write wins (optimistic locking via conditional DB update).
- Visitor exits from a gate different to entry gate: Allowed. Logged with cross-gate flag.
- Visitor with same name arrives simultaneously at two gates: Allowed; different records. Guard must compare selfie.
- Admin deletes a gate that has active visits: Gate cannot be deleted while active visits exist. Deactivate only.

- Selfie captured in very low light: Basic client-side brightness check warns visitor before submission.
- Form submitted with future expected exit time in the past: Validation error; must select a future time.
- Visitor submits from an unknown QR code (tampered URL): gate_id validation fails; form shows error.

Deliverable G: Dashboards & Metrics

- Live queue depth per gate (count, with trend last 1 hour).
- Average guard response time (submission to decision, per guard, per gate).
- Approval rate per gate / per guard (%).
- Active inside count (real-time, all gates combined and per-gate).
- Overstay count (current and historical trend).
- Category distribution (pie chart, today/week/month).
- Peak hour analysis (bar chart: submissions by hour of day).
- Gate load comparison (which gates are busiest).
- Rejection reasons breakdown (top 5 reasons).
- Guard shift workload (decisions per shift, per guard).

Deliverable J: Suggested Future AI/Automation Features

- Face Similarity Score: At gate, guard takes a second photo; AI compares to selfie and returns similarity %. Guard still decides.
- Anomaly Detection: Flag visitors with 3+ rejections across gates in 7 days for supervisor review.
- Gate Load Prediction: Predict next 2-hour visitor volume per gate based on time, day, and university events calendar.
- Smart Category Suggestion: Based on field entries (vehicle type, purpose), system suggests correct category if wrong one selected.
- Watchlist Expansion: Automatically flag any visitor whose description matches an existing watchlist entry (name + phone similarity).
- Document OCR: Scan and auto-fill form fields from photographed government ID.
- Behaviour Pattern Analysis: Identify suspicious patterns (same visitor, different names, multiple phones, multiple gates same day).

SECTION 44 — PRODUCT SUMMARY & RECOMMENDED BUILD ORDER

Product Summary

The Thapar University Visitor Management System is a multi-gate, role-layered, real-time digital platform that replaces paper gate registers with a structured, accountable, and searchable system. Visitors self-register via gate-specific QR codes on their mobile phones. Guards process entries from a live queue, verify selfies, and approve or reject with one tap. Exits are logged by any guard at any gate. Supervisors and the Head Admin gain cross-gate visibility, escalation handling, and full audit control. The system is designed for real gate conditions: fast, mobile-first, resilient, and operationally simple for guards while providing deep analytics and compliance controls for administration.

Recommended Build Order

70. Sprint 1–2: Foundation — Database schema (PostgreSQL), Auth service (JWT + RBAC), User management API, Gate setup.
71. Sprint 3–4: Visitor Form — All 12 category forms with dynamic fields, selfie capture, QR gate detection, submission API with deduplication.
72. Sprint 5–6: Guard Panel — Queue view (WebSocket), Visit detail view, Approve/Reject/Escalate actions, Entry logging.
73. Sprint 7: Exit Workflow — Active-inside search, cross-gate exit, Mark Exit API and UI.
74. Sprint 8: Supervisor Panel — Multi-gate view, Escalation handling, Override with audit trail.
75. Sprint 9: Admin Panel — Dashboard, User management UI, Audit trail view, Blacklist management.
76. Sprint 10: Alerting & Overstay — Background job, three-tier alert system, in-app notifications.
77. Sprint 11: Reports & CSV — Report engine, all report types, CSV export with queuing for large datasets.
78. Sprint 12: Hardening — Security audit (OWASP), Performance testing, Cross-browser testing, Documentation.
79. Sprint 13: Pilot — Deploy on staging, run with 1 gate, gather guard feedback, iterate.
80. Sprint 14: Full Go-Live — All 4 gates. Parallel run with paper registers for 2 weeks.

SECTION 45 — BEST PROMPT FOR CLAUDE CODE GENERATION

Use the following prompt when feeding this document into Claude Code to generate the application:

Build a complete full-stack Visitor Management System for Thapar University with the following spec:

TECH STACK: Node.js/Express backend, React + TailwindCSS frontend (two separate apps: visitor-form, and guard-admin panel), PostgreSQL database, Redis for sessions, Socket.IO for real-time queue updates.

SCOPE: (1) Public visitor form triggered by QR code with gate auto-detection. (2) Guard panel with live WebSocket queue, approve/reject/escalate, exit search and mark-exit. (3) Supervisor panel with multi-gate view and escalation handling. (4) Admin panel with user management, blacklist, audit trail, reports, and CSV export.

DATABASE: Create full PostgreSQL schema for: users (guard/supervisor/admin), gates (4), visitor_categories (12), visits (with full status lifecycle and form_data JSONB), guard_gate_assignments, blacklist, audit_logs, overstay_alerts.

AUTH: JWT-based auth, bcrypt password hashing, gate selection at login for guards, role-based middleware (guard/supervisor/head_admin).

REAL-TIME: Socket.IO room per gate. Guard panel subscribes to their gate room. New submission emits to gate room.

OVERSTAY JOB: Background cron every 5 minutes. Checks all entered/overstayed visits against expected_exit_at. Fires three-tier alerts.

STATUS LIFECYCLE: submitted → under_review → escalated/approved/rejected → entered → overstayed/exited/cancelled/expired.

KEY APIs: POST /api/visits/submit (public), GET /api/guard/queue (guard+), POST /api/guard/queue/:id/approve, GET /api/guard/active-inside, POST /api/guard/exit/:id, GET /api/supervisor/all-queues, POST /api/supervisor/escalations/:id/resolve, GET /api/admin/audit-trail, full CRUD for blacklist and users.

PHOTO: Compress selfie to max 800KB client-side before upload. Store in /uploads with visit_id-based filename. Return secure URL.

Generate: (a) Complete backend with all routes, middleware, and DB migrations. (b) Visitor form React app. (c) Guard/Admin React app. (d) Docker Compose file for local dev. (e) README with setup instructions. Use TypeScript where possible. Add input validation (Zod). Use environment variables for all secrets.

SECTION 46 — QUESTIONS STILL UNRESOLVED

The following questions require university administration decisions before or during development:

81. Official photo retention period: Has the university legal team reviewed and approved a specific data retention duration for selfies under DPDP Act 2023?
82. Who is the Head Admin? Is it the Chief Security Officer, Registrar, or IT Head? This affects email alerts and audit access.
83. Hostel list: Provide the complete and current list of all hostels on campus for the form dropdown.
84. Shift hours: Are there defined guard shift hours (e.g., 6AM–2PM, 2PM–10PM, 10PM–6AM)? Should the system support shift-based accountability reporting?
85. Gate operating hours: Are all 4 gates open 24/7, or do some gates close at night? System needs to handle 'gate closed' state.
86. Event frequency: How often are large-scale events (convocation, fests) held? This determines whether Event Mode is needed in Phase 1 or can wait.
87. Recurring contractors: Are there contractors who visit daily or weekly? Recurring Pass system priority depends on this answer.
88. SMS/WhatsApp: Is there a budget for SMS gateway API integration for visitor notifications?
89. Network infrastructure: Is there a dedicated network at each gate, or are guards using personal mobile data? This affects WebSocket reliability design.
90. Student phone number: Should the Parent visitor form make student contact number mandatory? This significantly aids verification but may create privacy concerns.

91. Vehicle entry restrictions: Are there specific vehicle type restrictions (e.g., no trucks at Front Gate)? System can enforce these if defined.
92. Data sovereignty: Must all data (including photos) remain on university premises (on-premise storage only), or is cloud storage permissible?